

VICKEY GHIMIRE

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Education

M.S. in Geospatial Information Science

The University of Texas at Dallas

Expected December 2027

Richardson, TX

B.S. in Computer Science

Northern Kentucky University | GPA: **3.94**/4.0

08/2022 - 05/2026

Highland Heights, KY

Technical Skills

Geospatial	ArcGIS Pro, ArcGIS, QGIS, GeoPandas, Rasterio, rioxarray, Xarray, Spatial SQL, Raster Analysis, DEM/Terrain Analysis, OSM, OSRM
Programming	Python, R, C++, Java, SQL, JavaScript
Data & ML	Pandas, NumPy, PyTorch, SciPy, scikit-learn, Time-Series Modeling, NLP
Web & Cloud	Django, FastAPI, PostgreSQL, REST APIs, AWS (ECS, RDS), Firebase

Experience

High Performance Computing Student Assistant

08/2026 – Present

High Performance Computing (HPC), UTD

Richardson, TX

- Provide technical support to researchers and students using HPC systems, troubleshooting Linux/UNIX environments, software compilation/installations, and scientific computing applications in coordination with HPC staff.
- Monitor HPC systems and usage data, assisting with system reporting, data visualization, and user support while contributing to technical workshops and research-computing training activities.

Full-Stack Django Developer

09/2025 – 05/2026

Department of Physics, NKU

Highland Heights, KY

- Developed a modular Django-based web application, refactoring a legacy script into an MTV-compliant architecture, improving code maintainability and enabling faster feature development.
- Built and integrated spatial data pipelines using OpenStreetMap and NASA POWER APIs to replace simulated inputs with site-level environmental data, improving accuracy of optimization workflows.
- Optimized a genetic-algorithm optimization workflow with convergence-based termination and result caching, cutting average computation time by **35%**.

Research Assistant- Data Analytics

10/2024 – 08/2026

Human-Computer Interaction Lab, NKU

Highland Heights, KY

- Designed an end-to-end novelty + relevance detection framework for LLM-generated content using hybrid semantic similarity (Jaccard + BERT), enabling automated content review of 9,000+ topics with **90%+** coverage.
- Led analysis of ChatGPT's memory feature across **135+** users, building reproducible pipelines for multi-account data cleaning, transformation, automated visualization, and response analysis.

Projects

Dallas Traffic Safety & Crash Hotspot Analysis | *ArcGIS Pro, Pandas, Spatial Analysis, Python* September 2026*

- Built a reproducible Python preprocessing pipeline for **160K+** Dallas traffic crash records (2021–2025), standardizing temporal, location, roadway, and KABCO injury-severity attributes & validating **149k+** geocoded records for analysis.
- Analyzed citywide crash and severe-injury patterns in ArcGIS Pro using spatial aggregation, Kernel Density, and Gi* hotspot analysis to identify statistically significant high-crash areas and priority transportation corridors.
- Evaluating spatial and temporal patterns in fatal and serious-injury crashes across roadway, intersection, time-of-day, and environmental conditions, with planned comparison against Dallas Vision Zero's High Injury Network.

Site Suitability Analysis: Fire Station in Kathmandu | *GeoPandas, xarray, rioxarray, rasterio* | [Link] July 2026

- Built a multi-criteria site-suitability pipeline by aligning OSM vector data and SRTM DEM rasters, deriving slope and Euclidean-distance to quantify terrain safety and emergency accessibility, to support operation and faster response time.
- Applied Weighted Linear Combination to normalized suitability scores, ranking candidate fire-station locations and exporting results to an interactive Leaflet map for ground evaluation.

Disaster-Aware Road Risk Viewer | *GeoPandas, React-Leaflet, Shapely, FastAPI* | [Link]

March 2026

- Developed a geospatial decision-support platform visualizing seasonal landslide/flood risk in Nepal's Bagmati Province.
- Segmented OSM road networks and derived slope and Terrain Ruggedness Index features to score risk, then matched OSRM routes to pre-scored segments for comparative route-safety analysis for possible travel paths.